

Agenda

- Modular Arithmetic Introduction ✓
- Count pairs whose sum mod m is 0 ✓
- Introduction to GCD ✓
- Properties of GCD ✓
- Question: Delete one

$A \% B \Rightarrow$ Remainder when A / B .

$$\underline{A \% B} \Rightarrow \underline{[0, B-1]}$$

$$\begin{array}{c} 14 \% 7 \\ \downarrow \\ 0 \end{array}$$

$$\begin{array}{c} 15 \% 2 \\ \downarrow \\ 1 \end{array}$$

$$\begin{array}{c} 5 \% 10 \\ \downarrow \\ 5 \end{array}$$

$$\begin{array}{c} 9 \% 10 \\ \downarrow \\ 9 \end{array}$$

$$\begin{array}{c} 0 \% 10 \\ \downarrow \\ 0 \end{array}$$

Why do we need Modulo operator?

→ Integer → no stack overflow, out of range.
long $\Rightarrow$ ground your number within the range.

$$\begin{array}{c} a+b \\ \underline{\quad} \end{array} = \underbrace{10^{10}} + \underbrace{10^{10}} \Rightarrow \text{integer}$$

→ $(ans) \% (10^9)$
↓
Store this.

Rules of applying Modulo operator:

$$(1) \quad (a+b) \% m = (a \% m + b \% m) \% m$$

$$a=9 \quad b=8 \quad m=5$$

$$(a+b) \% m = (9+8) \% m = 17 \% 5 = \underline{2} \text{ ans.}$$

$$(9 \% 5 + 8 \% 5) \% 5$$

$$(4 + 3) \% 5 = 7 \% 5 = \underline{2} \text{ ans.}$$

$$(2) \quad (a * b) \% m = (a \% m * b \% m) \% m$$

$$(3) \quad (a + m) \% m = (a \% m + \underbrace{m \% m}_0) \% m \\ = (a \% m) \% m \\ = a \% m$$

$$a=7 \quad m=3$$

$$a \% m = \underbrace{7}_{< m} \% m = \underline{\underline{7}}$$

$$(4) \quad (a-b) \% m = (a \% m - b \% m) \% m$$

$$a=7 \quad b=10 \quad m=5$$

$$(a-b) \% m = (7-10) \% 5$$

$$= \underbrace{-3}_{> 0} \% 5$$

$$\underbrace{-3 + 5}_{= 2} = \underline{\underline{2}}$$

$$\Rightarrow (a-b) \% m = (a \% m - b \% m + m) \% m$$

$$a = 7 \quad b = 10 \quad m = 5$$

$$\begin{aligned} & (7 \% 5 - 10 \% 5 + 5) \% 5 \\ &= (2 - 0 + 5) \% 5 \\ &= 7 \% 5 = \underline{\underline{2}} \end{aligned}$$

$$\textcircled{5} = (a^b) \% m = ((a \% m)^b) \% m$$

$$a^b$$

$$\underbrace{(a * a * a \dots a)}_{b \text{ times}} \% m$$

$$\Rightarrow (a \% m * a \% m * a \% m \dots a \% m) \% m$$

$$\Rightarrow ((a \% m)^b) \% m$$

Quiz

$$(37^{10^3} - 1) \% 12$$

$$(a - b) \% 12$$

$$(a \% 12 - b \% 12 + 12) \% 12$$

$$\Rightarrow (37^{10^3} \% 12 - 1 \% 12 + 12) \% 12$$

$$\Rightarrow ((37 \% 12)^{10^3}) \% 12 - 1 + 12) \% 12$$

$$\begin{aligned}
 &= \left(\underbrace{1^{103} \% 12}_{\downarrow} - 1 + 12 \right) \% 12 \\
 &= (\cancel{1} - \cancel{1} + 12) \% 12 \\
 &= 0 \quad \underline{\underline{\text{ans}}}
 \end{aligned}$$

Ques: Given N array elements. find count of pair (i, j) such that $(a[i] + a[j]) \% m = 0$ where $i \neq j$ and $\text{pair}(i, j) = \text{pair}(j, i)$

$$A[] = [4, 3, 6, 3, 8, 12]$$

$$m = 6$$

$$(4 + 2) \% 6 = 0$$

$$(3 + 3) \% 6 = 0$$

$$(6 + 12) \% 6 = 0$$

Brute Force

- ✓ ① → Find all the pairs
- ✓ ② check $(a + b) \% m == 0$ ✓

$$\underline{\underline{\text{T.C}}}: O(n^2)$$

Optimized Approach

$$A = [2, 3, 4, 8, 6, 15, 5, 12, 17, 7, 18]$$

$$A \% m$$

$$A \% 6 = [2, 3, 4, 2, 0, 3, 5, 0, 5, 1, 0]$$

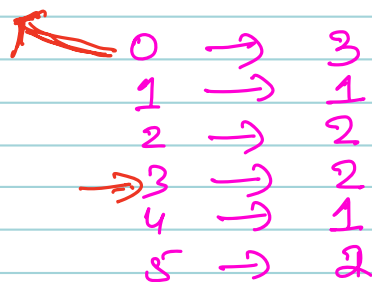
$$(a + b) \% m = (a \% m + b \% m) \% m = 0$$

$$[0 \quad m-1] \quad [0, m-1]$$

$$1 \quad m-1$$

$$2 \quad m-2$$

$$3 \quad m-3$$



$$m = 6$$

$$a = 1$$

$$b = 5$$

$$m-1$$

$$1$$

$$2$$

$$= 2 \text{ pairs}$$

m is even.

$$\boxed{m/2} =$$

$$(m + 0) \% m = 0$$

$$(0 + 0) \% m = 0$$

$$\text{if } n \text{ digit} = \text{pairs} = \frac{n \times (n-1)}{2}$$

even

$$m = 6$$

$$(3 + 3) \% 6 = 0$$

→ handle : 0 and even m cases.

$$\Rightarrow \frac{\text{freq}(0) * \text{freq}(0) - 1}{2}$$

$$m \% 2 == 0$$

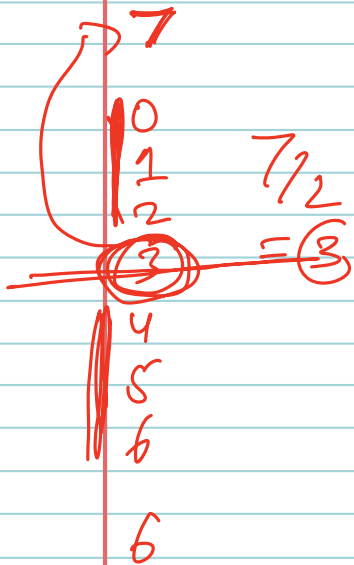
$$\checkmark = \frac{\text{freq}(m/2) * (\text{freq}(m/2) - 1)}{2}$$

pseudo

$$\Rightarrow \text{hashmap} = [\text{arr}] \% m$$

$$\text{count} = 0$$

$$\text{count} += \frac{\text{hashmap}[0] * \text{hashmap}[0] - 1}{2}$$



$$\text{if } m \% 2 == 0$$

$$\text{count} += \frac{\text{hashmap}[m/2] * \text{hashmap}[m/2] - 1}{2}$$

$$\text{for (int } i = 1; i < \frac{m+1}{2}; i++)$$

$$\text{count} += \text{hashmap}[i] * \text{hashmap}[m-i]$$

Keys hashmap = m

$$\underline{\text{S.C}} = O(m)$$

10:30 pm

GCD

↳ Greatest common divisor.

HCF = Highest common factor.

$$\text{GCD} (15, 25) = 5$$

↓	↓
1	1
3	5
5	25
15	

$$\text{GCD} (12, 30) = 6.$$

↓	↓
1	1
2	2
3	3
4	5
6	6
12	10
	15
	30

$$\text{GCD}(0, 4) = 4$$

↓	↓
1	1
2	2
3	4
4	
5	
⋮	
∞	

Any number
gcd with 0 is
that number.

$$\text{GCD}(0, 0) \Rightarrow \text{empty.}$$

1	1
2	2
3	3
⋮	⋮
0	0

Properties of GCD.

$$(1) \quad \text{GCD}(A, B) = \text{GCD}(B, A)$$

$$(2) \quad \text{GCD}(0, A) = A \quad \checkmark$$

$$(3) \quad \begin{aligned} \text{GCD}(A, B, C) &= \text{GCD}(A, \text{GCD}(B, C)) \\ &= \text{GCD}(B, \text{GCD}(A, C)) \\ &= \text{GCD}(C, \text{GCD}(A, B)). \end{aligned}$$

$$A = 2 \quad B = 3 \quad C = 4$$

$$\begin{aligned} \text{GCD}(A, \text{GCD}(B, C)) &= \text{GCD}(2, \text{GCD}(3, 4)) \\ &= \text{GCD}(2, 1) \\ &= 1 \end{aligned}$$

$$\begin{aligned} \text{GCD}(B, \text{GCD}(A, C)) &= \text{GCD}(3, \text{GCD}(2, 4)) \\ &= \text{GCD}(3, 2) \\ &= 1 \end{aligned}$$

④ Given $A \geq B > 0$

$$\text{GCD}(A, B) = \text{GCD}(A - B, B)$$

$$\begin{aligned} \text{GCD}(5, 3) \\ = 1 \end{aligned}$$

$$A = 5 \quad B = 3$$

$$\begin{aligned} \text{GCD}(2, 3) \\ \downarrow \\ 1 \end{aligned}$$

$$A = 10$$

$$B = 6$$

$$\begin{aligned} \text{GCD}(10, 6) \\ \downarrow \\ 2 \end{aligned}$$

$$\begin{aligned} \text{GCD}(4, 6) \\ \downarrow \\ 2 \end{aligned}$$

⑤ *** $\text{GCD}(A, B) = \text{GCD}(A \% B, B)$

$A = 20 \quad B = 12$

$\text{GCD}(20, 12)$

$\downarrow$
4

$\text{GCD}(20 \% 12, 12)$

$\downarrow$
 $\text{GCD}(8, 12)$
 $\downarrow$
4

$\text{GCD}(A, B) = \text{GCD}(B, A \% B)$

$\text{GCD}(14, 21) \Rightarrow \text{GCD}(21, 14) \xrightarrow{21 \% 14 = 7} \text{GCD}(14, 7)$
 $A \quad B$
 $A \% B$

$\downarrow$
 $\text{GCD}(7, 0)$
 $\downarrow$
7

$\text{GCD}(21, 14)$

not optimal.

$\text{GCD}(A, B) = \text{GCD}(A - B, B) \quad A > B$

$\text{GCD}(21, 14) = \text{GCD}(7, 14) \Leftrightarrow \text{GCD}(14, 7)$

$\downarrow$
 $\text{GCD}(7, 7) \rightarrow \text{ans} = 7$
 $\downarrow$
 $\text{GCD}(0, 7) \rightarrow$

GCD

T.C $\log(n)$

```
int gcd (a, b) {  
    if (b == 0) return a;  
    return gcd (b, a % b);  
}
```

$[0, 0] \rightarrow \underline{\text{infity}}$

GCD

Ques

Find the GCD of entire array

A = [6, 12, 15] ans = 3

```
graph TD  
    A["A = [ 6, 12, 15 ]"] --> B["ans"]  
    B --> C["ans"]
```

```
int gcd arr (Arr) {
```

```
    int ans = arr[0]
```

```
    for ( i=1; i < arr.length; i++ )
```

```
    {
```

```
        ans = gcd (ans, arr[i]);
```

```
    }
```

```
    return ans;
```

T.C

$O(N * \log N)$

Ques Given arr[N] element.

We have to delete one element such that the GCD of remaining elements become maximum.

$$\text{arr} = \begin{matrix} 0 & 1 & 2 & 3 & 4 \\ [24, & 16, & 18, & 30, & 15] \end{matrix}$$

$$\begin{matrix} 0 & 1 & 2 & 3 & 4 \\ [\cancel{24}, & 16, & 18, & 30, & 15] \end{matrix} = 1$$

$$\begin{matrix} 0 & 1 & 2 & 3 & 4 \\ [24, & \cancel{16}, & 18, & 30, & 15] \end{matrix} = 3 \quad (16)$$

$$\begin{matrix} 0 & 1 & 2 & 3 & 4 \\ [24, & 16, & \cancel{18}, & 30, & 15] \end{matrix} = 1$$

$$\begin{matrix} 0 & 1 & 2 & 3 & 4 \\ [24, & 16, & 18, & \cancel{30}, & 15] \end{matrix} = 1$$

$$\begin{matrix} 0 & 1 & 2 & 3 & 4 \\ [24, & 16, & 18, & 30, & \cancel{15}] \end{matrix} = 2$$

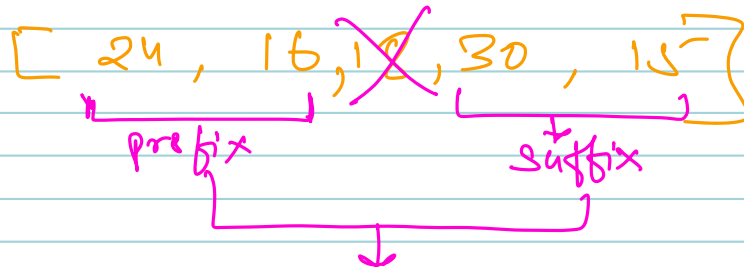
Brute Force

- ① Run a loop and skip 1 index each time
- ② Calculate GCD of remaining elements
- ③ $\rightarrow$ Find max.

T.C $O(N^2 \log(N))$

Optimize

prefix GCD and Suffix GCD.
GCD.



TC Prefix: $O(N \log N)$ GCD
Suffix: $O(N \log N)$ $\downarrow$
 $O(N) \rightarrow$ iterating. ans

TC $n \log n$

$[24, 16, 18, 30, 15]$

$\rightarrow$ suffix array

$S.C = O(n)$

Pre/Suffix array

$[24, 16, 18, 30, 15]$
 $\downarrow$
Prefix array.

H.W